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PLASMA-FREE ETCHED B-GA₂O₃-NIO MERGED PIN SCHOTTKY DIODE WITH HIGH-VOLTAGE STRESS RELIABILITY

Abstract

A (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device and method of making the same are presented. In implementations, the method includes: providing a Ga.sub.2O.sub.3 base including a Ga.sub.2O.sub.3 substrate with an n-type Ga.sub.2O.sub.3 epitaxial layer thereon; forming NiO-filled internal trenches and an NiO-filled peripheral trench in the n-type Ga.sub.2O.sub.3 epitaxial layer using a plasma-free etching process; forming at least one junction termination extension (JTE) structure about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer; depositing an anode over the NiO filled interior trenches, over portions of the NiO-filled peripheral trench, and over portions of the JTE structure; and depositing a cathode over a bottom surface of the Ga.sub.2O.sub.3 substrate, thereby forming the Ga.sub.2O.sub.3—NiO heterojunction device, wherein the Ga.sub.2O.sub.3—NiO heterojunction device is formed without the use of plasma-etching and is free of plasma-etching damage.

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Background/Summary

BACKGROUND OF THE INVENTION

[0002] Aspects of the present invention relate generally to semiconductor devices and, more particularly, to a plasma-etch-free method of making a gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device.

[0003] The stable beta (β) phase of Ga.sub.2O.sub.3 has received significant interest due to its scalable melt growth and properties favorable for high power electronics, such as an ultra-wide bandgap (UWBG) of 4.8 electron volt (eV), controllable donor doping across a wide range (10.sup.14-10.sup.21 cm.sup.-3), and a high critical breakdown electric field (6-8 megavolts). Ga.sub.2O.sub.3 is also the only wide bandgap (WBG) or UWBG semiconductor that can be grown from a melt; offering potential cost advantages over silicon carbide (SiC) and gallium nitride (GaN). Substrate and epitaxial growth technology for Ga.sub.2O.sub.3 is advancing rapidly, and 100 millimeter (mm) epitaxial wafers (epiwafers) are currently commercially available. As the field of high-power electronics continues to expand, Ga.sub.2O.sub.3 has shown promise as a candidate semiconductor for the next generation of power electronics.

[0004] The advantages mentioned above make Ga.sub.2O.sub.3 a candidate for vertical power rectifiers such as Schottky barrier (SBD) and trench Schottky diodes. These diodes are unipolar devices utilizing high quality n.sup.- drift layers. On the other hand, PN diodes, junction barrier Schottky (JBS) diodes, and merged PiN Schottky (MPS) diodes are bi-polar devices relying on both electrons and holes to carry charge across an abrupt PN junction. This poses an issue for Ga.sub.2O.sub.3 where p-type conductivity has yet to be realized by means of either growth or implantation of acceptor species.

[0005] One approach to realize Ga.sub.2O.sub.3 bi-polar devices proposes the use of anisotype heterojunctions. Sputtered p-type NiO with a band gap of 3.7 eV and a favorable band offset to Ga.sub.2O.sub.3 has thus far been the most researched p-type semiconductor due to ease of deposition and tunable acceptor concentration. Ga.sub.2O.sub.3—NiO PN diodes are fabricated by sputtering NiO atop a planar Ga.sub.2O.sub.3 drift layer, while the Ga.sub.2O.sub.3 NiO JBS and MPS diodes, identical in architecture, are further complicated by the need to embed the array of heterojunctions into the drift layer using various etch techniques. More mature, bi-polar WBG semiconductors such as SiC and GaN are not plagued with this issue as selective p-type implantation has been demonstrated with aluminum (Al) and magnesium (Mg), respectively. Patterned formation of Ga.sub.2O.sub.3—NiO heterojunctions introduces a host of issues yet to be addressed in the literature. Many works have studied Ga.sub.2O.sub.3 plasma etch damage, post etch treatments for damage recovery, and their impact on device performance. However, this issue is not quantified in studies involving etched Ga.sub.2O.sub.3—NiO heterojunctions. Recently, a

plasma-free low-damage Ga.sub.2O.sub.3 etching technique using atomic Ga flux was demonstrated and applied in Ga.sub.2O.sub.3 trench Schottky barrier diodes. Compared to a Schottky diode, a junction-barrier Schottky (JBS) or a merged p-i-n Schottky (MPS) diode can offer reduced reverse leakage current and increased breakdown voltage, which makes these devices more practical from a commercialization standpoint.

[0006] Accordingly, there remains a need for Ga.sub.2O.sub.3 bi-polar devices, also known as bipolar junction transistors. In implementations of the invention discussed herein, we have demonstrated that the combination of a plasma-free Ga.sub.2O.sub.3 etch, a p-type NiO layer, and a high work function metal oxide layer (e.g., PtOx), which acts also as an Ohmic contact to the p-NiO layer, can result in improvement of both forward and reverse bias characteristics of a rectifier.

SUMMARY OF THE INVENTION

[0007] In a first aspect of the invention, there is a method of making a gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device including: providing a Ga.sub.2O.sub.3 base including a Ga.sub.2O.sub.3 substrate with an n-type Ga.sub.2O.sub.3 epitaxial layer thereon; forming NiO-filled internal trenches and an NiO-filled peripheral trench in the n-type Ga.sub.2O.sub.3 epitaxial layer using a plasma-free etching process; forming at least one junction termination extension (JTE) structure about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer; depositing an anode over the NiO filled interior trenches, over portions of the NiO-filled peripheral trench, and over portions of the JTE structure; and depositing a cathode over a bottom surface of the Ga.sub.2O.sub.3 substrate, thereby forming the Ga.sub.2O.sub.3—NiO heterojunction device, wherein the Ga.sub.2O.sub.3—NiO heterojunction device is formed without the use of plasma-etching and is free of plasma-etching damage. In embodiments, internal trenches in the n-type Ga.sub.2O.sub.3 epitaxial layer are formed via the plasma-free etching process; a peripheral trench is formed about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer via the plasma-free etching process; and p-type NiO is deposited into the internal trenches and the peripheral trench, thereby forming the NiO-filled internal trenches and an NiO-filled peripheral trench.

[0008] In implementations, forming the JTE structure includes: depositing the at least one JTE structure over portions of the NiO filled peripheral trench and surface portions of the Ga.sub.2O.sub.3 epitaxial layer. In embodiments, the Ga.sub.2O.sub.3—NiO heterojunction device is a merged PiN Schottky (MPS) diode having a specific contact resistance between the anode and the NiO of less than 0.0005 Ohm*cm.sup.2. In aspects of the invention, the JTE structure comprises a bi-layer JTE structure including first and second layers of NiO. In implementations, depositing the JTE structure includes: depositing a first layer of NiO via sputtering in a pure argon atmosphere over portions of the peripheral NiO-filled trench and over portions of a surface of the Ga.sub.2O.sub.3 epitaxial layer; depositing a first stage of a second layer of NiO in 20:1 argon:oxygen atmosphere over the first layer and portions of the peripheral NiO-filled trench; and depositing a second stage of the second layer of NiO in a 2:1 argon:oxygen atmosphere over the first stage of the second layer of NiO and portions of the peripheral NiO-filled trench. In embodiments, the JTE structure comprises a layer of NiO. In implementations, forming the NiO-filled internal trenches and the NiO-filled peripheral trench in the n-type Ga.sub.2O.sub.3 epitaxial layer utilizes a plasma-free etching process comprises patterning using photolithography and gallium flux etching. The method may further include annealing the Ga.sub.2O.sub.3—NiO heterojunction device. The anode and cathode may comprise platinum oxide and gold. In implementations, the n-type Ga.sub.2O.sub.3 epitaxial layer is doped with silicon (Si).

[0009] In another aspect of the invention, there is a gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device including: a Ga.sub.2O.sub.3 base comprised of a Ga.sub.2O.sub.3 substrate with an n-type Ga.sub.2O.sub.3 epitaxial layer thereon, the n-type Ga.sub.2O.sub.3 epitaxial layer including a plurality of NiO-filled internal trenches and an NiO-filled peripheral trench; a junction termination extension (JTE) structure formed about the periphery of the n-type

Ga.sub.2O.sub.3 epitaxial layer; an anode formed over the NiO filled interior trenches, over portions of the NiO-filled peripheral trench, and over portions of the JTE structure; and a cathode formed over a bottom surface of the Ga.sub.2O.sub.3 substrate, wherein the Ga.sub.2O.sub.3—NiO heterojunction device is formed without the use of plasma-etching and is free of plasma-etching damage. In implementations, the JTE structure is a NiO bi-layer JTE structure. In embodiments, the first layer of the NiO bi-layer JTE structure extends beyond opposing edges of the anode. In aspects of the invention, the JTE structure comprises a layer of NiO. The anode and cathode may be made of platinum oxide and gold.

[0010] In embodiments, the JTE structure is formed over portions of the NiO filled peripheral trench and surface portions of the Ga.sub.2O.sub.3 epitaxial layer. In aspects of the invention, the Ga.sub.2O.sub.3—NiO heterojunction device is a merged PiN Schottky (MPS) diode having a specific contact resistance between the anode and the NiO of less than 0.0005 Ohm*cm.sup.2. In embodiments, the n-type Ga.sub.2O.sub.3 epitaxial layer is doped with silicon (Si). In some implementations, the n-type Ga.sub.2O.sub.3 epitaxial layer has a thickness of at least 5 microns.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Aspects of the present invention are described in the detailed description which follows, in reference to the noted plurality of drawings by way of non-limiting examples of exemplary embodiments of the present invention.

[0012] FIG. 1 shows a flowchart of an exemplary method for making a semiconductor device in accordance with aspects of the present invention.

[0013] FIG. 2A depicts a stage of a device of the present invention including a SiO.sub.2-patterned mask with interior trenches and a peripheral trench formed therein.

[0014] FIG. 2B depicts a stage of a device of the present invention after Ga flux etching.

[0015] FIG. 2C depicts a stage of a device of the present invention after the residual portions of the SiO.sub.2 mask are removed.

[0016] FIG. 2D depicts a stage of a device of the present invention with NiO-filled interior trenches and an NiO-filled peripheral trench.

[0017] FIG. 2E depicts a Ga.sub.2O.sub.3—NiO heterojunction device of the present invention with opposing NiO JTEs portions and an anode and cathode.

[0018] FIG. 2F is a close-up partial view of FIG. 2E, showing the two stages of the second bi-layer of the NiO JTE structure.

[0019] FIG. 2G depicts another Ga.sub.2O.sub.3—NiO heterojunction device of the present invention with opposing NiO JTE portions and an anode and cathode.

[0020] FIG. 2H is a top view of the embodiment 200G of FIG. 2G.

[0021] FIG. 3A is a chart depicting C-V measurements of a PtOx and Ni SBD in accordance with embodiments of the invention

[0022] FIG. 3B is a chart depicting C-V measurements of a PtOx JBS diode in accordance with embodiments of the invention, showing a dual sweep at 300 kHz measured from accumulation to depletion (dashed line) and back to accumulation (solid line).

[0023] FIG. 3C is a chart depicting linear transmission line measurements of Ni and PtOx Ohmic contacts to NiO. Control films of NiO were isolated via lift-off on sapphire substrates, as indicated in FIG. 3C.

[0024] FIG. 4A is a chart depicting C-V measurements for a PtOx MPS and Ni JBS diodes in accordance with embodiments of the invention.

[0025] FIG. 4B is a chart depicting C-V measurements Ni/Au and PtOx/Au SBDs in accordance with embodiments of the invention, showing a dual sweep at 300 kHz measured from accumulation

to depletion and back to accumulation.

[0026] FIG. 5 illustrates a testing waveform for high voltage reliability measurements in accordance with embodiments of the invention.

[0027] FIG. 6 is a chart depicting specific on-resistance and reverse leakage current as a function of stress time measured at 600, 700 and 800 V.

[0028] FIG. 7 is a chart depicting forward J-V measurements (25 curves) taken during the reliability measurements of the MPS diode at 700V.

DETAILED DESCRIPTION

[0029] Aspects of the present invention relate generally to semiconductor devices and, more particularly, to a plasma-free etching method of making a gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device. In embodiments, a method of fabricating a beta-gallium oxide ($\beta\text{-Ga.sub.2O.sub.3}$)—NiO merged PiN Schottky (MPS) diode is provided, which uses a gallium (Ga) flux plasma-free etch process and platinum oxide (PtOx) contacts. In general, an MPS diode is a hybrid device that combines a Schottky diode and a P-N diode in parallel.

[0030] Dry etching or plasma etching is a process of removing material, typically a masked pattern of semiconductor material, by exposing the material to a bombardment of ions (e.g., a plasma of nitrogen, chlorine, and boron trichloride), that dislodge particles of the material from the exposed surface. One drawback of plasma etching is the problem of surface damage that may result from the etching process, which can hinder device performance by increasing the level of leakage current (i.e., the amount of current that flows when it should not). For example, a high power (typically 600-1000 W) boron trichloride (BCl.sub.3) reactive-ion etching (RIE) or inductively-coupled plasma (ICP) etching has long been known to negatively impact Ga.sub.2O.sub.3 surfaces and requires mitigation with high temperature or aggressive chemical treatments. While instances of plasma damage to Ga.sub.2O.sub.3 surfaces and Schottky contact formations are readily found within the literature, the impacts on a Ga.sub.2O.sub.3 NiO etched heterojunction remain a void.

[0031] Advantageously, the use of a plasma-free etch process in implementations of the invention avoids the problem of Ga.sub.2O.sub.3 surface damage that would be caused by a high-power plasma etch process. By removing the need for RIE plasma etching and post etch chemical treatments, embodiments of the invention provide a high quality and damage free surface for heterojunction formation.

[0032] In embodiments, PtO.sub.x acts as both a Schottky contact to n-type Ga.sub.2O.sub.3 as well as an Ohmic contact to p-type NiO. Compared to Ni contacts to NiO (Ni/NiO contacts) used in Ga.sub.2O.sub.3 junction barrier Schottky (JBS) diodes and PN diodes, the $\text{PtO.sub.x}/\text{NiO}$ contact in accordance with embodiments of the invention exhibits a lower contact resistance as shown by linear transmission line measurements. This improved contact resistance enables an apparent second turn-on, potentially due to minority carrier injection, and a further decrease in differential on-resistance under forward diode operation, confirming the formation of a Ga.sub.2O.sub.3 MPS diode.

[0033] MPS diodes fabricated in accordance with embodiments of the invention were subjected to high reverse bias reliability testing. At 800 volts (V), MPS diodes were stressed for a cumulative time of 2000 seconds(s) while periodically switching the device on. No variation or degradation of the on-state performance was observed as the Schottky regime (<3 V) of the MPS diode was turned on. This signifies an improved Ga.sub.2O.sub.3 NiO heterojunction with minimal side-wall trapping as a result of the plasma-etch-free process.

Exemplary Method

[0034] FIG. 1 shows a flowchart of an exemplary method for making a semiconductor device in accordance with aspects of the present invention. The method references semiconductor device elements depicted in FIGS. 2A-2F.

[0035] At **101**, a gallium oxide (Ga.sub.2O.sub.3) base **202** is provided or fabricated, wherein the Ga.sub.2O.sub.3 base **202** is comprised of an n.sup.+ beta (β)-phase Ga.sub.2O.sub.3 substrate **203**

with an n-Ga.sub.2O.sub.3 epitaxial layer **204** grown thereon. See FIG. 2A described below. In embodiments, the substrate **203** is a tin (Sn)-doped β -Ga.sub.2O.sub.3 (β -Ga.sub.2O.sub.3:Sn) substrate with a halide vapor phase epitaxial (HVPE) layer **204** thereon. In implementations, the β -Ga.sub.2O.sub.3:Sn substrate has a thickness of 5 micrometers (μm), and includes unintentional silicon (Si) doping.

[0036] At **102**, multiple internal trenches (e.g., **210A**, **210B**, **210C**) and at least one peripheral trench **215** (hereafter the peripheral trench) are formed in the n-type Ga.sub.2O.sub.3 epitaxial layer **204** via a plasma-free anisotropic dry etching method. In embodiments, the trenches are formed using photolithographic gallium flux etching. In implementations, the peripheral trench **215** extends about the periphery of the Ga.sub.2O.sub.3 epitaxial layer **204**. Opposing portions **215A** and **215B** of a single peripheral trench **215** are depicted in FIG. 2D discussed below, for example.

[0037] In one exemplary embodiment, a silicon oxide (SiO.sub.2) mask layer **206** is deposited onto a surface of the Ga.sub.2O.sub.3 epitaxial layer **204** using plasma enhanced chemical vapor deposition (PECVD). Existing PECVD methods may be utilized to deposit the SiO.sub.2 mask layer **206**. The SiO.sub.2 mask layer **206** is then patterned using a buffered oxide etchant (BOE), thereby generating a SiO.sub.2-patterned surface of the Ga.sub.2O.sub.3 epitaxial layer **204**.

Existing patterning methods may be utilized to implement step **103**. In embodiments, the patterning results in interior trenches **208A-208C** formed in the SiO.sub.2 mask, as well as the peripheral trench **215**. See FIG. 2A discussed below. The SiO.sub.2-patterned surface of the Ga.sub.2O.sub.3 base **202** is heat-cleaned. In implementations, the heat cleaning is performed for ten (10) minutes at 800 degrees Celsius ($^{\circ}\text{C}$.). Existing heat cleaning methods may be utilized. See FIG. 2A discussed below.

[0038] FIG. 2A is a cross-sectional side view of a stage **200A** of a device of the present invention including a SiO.sub.2-patterned mask **206**, with the interior trenches **208A-208C** and a peripheral trench **209** formed therein. Opposing portions of the peripheral trench **209** are shown at **209A** and **209B**. The device stage **200A** includes the Ga.sub.2O.sub.3 base **202** comprised of the n+ beta (β)-phase Ga.sub.2O.sub.3 substrate **203** with the n.sup.- Ga.sub.2O.sub.3 epitaxial layer **204** grown thereon. While three interior trenches **208A-208C** are depicted, it should be understood that more or less interior trenches may be formed in accordance with embodiments of the invention.

[0039] As part of the exemplary photolithography process of step **102**, Ga flux etching of the SiO.sub.2-patterned surface **206** of the Ga.sub.2O.sub.3 base is performed, resulting in an etched n-Ga.sub.2O.sub.3 epitaxial layer **204**, including interior trenches **210A-210C**, and a peripheral trench **211** (opposing sides of which are shown at **211A** and **211B**). See FIG. 2B discussed below. In general, Ga flux etching is a technique that uses Ga to etch the surface of Ga.sub.2O.sub.3 and other materials. Ga flux etching uses molecular beam epitaxy (MBE) and other processes known to those skilled in the art. In implementations, the Ga flux etching is performed for 159 minutes (min) using $\sim 1.24 \times 10^{-7}$ Torr of Ga flux at 725°C . thermocouple temperature in an MBE reactor maintained at 1×10^{-9} Torr during the etching.

[0040] FIG. 2B is a cross-sectional side view of a stage **200B** of a device of the present invention after the Ga flux etching. Ga flux etching results in the interior trenches **210A-210C** and the peripheral trench **211** (with opposing sides indicated at **211A** and **211B**) formed in the Ga.sub.2O.sub.3 epitaxial layer of the Ga.sub.2O.sub.3 base **202**. While three interior trenches **211A-211C** are depicted, it should be understood that more or less interior trenches may be formed in accordance with embodiments of the invention. Ga flux etching may result in residual droplets of Ga on surfaces of the device, as represented at **212**.

[0041] As part of the exemplary photolithography process of step **102**, portions of the SiO.sub.2 mask **206** that remain after the Ga flux etching step are removed using BOE, resulting in a patterned Ga.sub.2O.sub.3 base **202** with trenches **210A-210C**, and **211** formed therein. See FIG. 2C discussed below.

[0042] FIG. 2C is a cross-sectional side view of a stage **200C** of a device of the present invention

after the residual portions of SiO₂ mask **206** are removed.

[0043] As part of the exemplary photolithography process of step **102**, the patterned Ga.sub.2O.sub.3 base **202** is soaked in hydrogen chloride (HCl) diluted in water (H₂O), to remove any remaining Ga droplets from surfaces of the Ga.sub.2O.sub.3 base **202**. In implementations, the dilution ratio of H₂O to HCl utilized is 10:1. The patterned Ga.sub.2O.sub.3 base **202** is re-patterned with another photoresist, resulting in a repatterned Ga.sub.2O.sub.3 base **202** including trenches **210A-210C**, **211A** and **211B** formed therein.

[0044] At **103**, nickel oxide (NiO) (a p-type material) is deposited into trenches **210A-210B** and **211** of the repatterned Ga.sub.2O.sub.3 base **202**, thereby creating NiO trench regions **214A-214C**, **215A** and **215B** within the repatterned Ga.sub.2O.sub.3 base **202**. See FIG. 2D discussed below. In implementations, the NiO is deposited via sputtering from an NiO target, with a hole concentration (N_A) of N_A>10¹⁸ cm⁻³. The photoresist may then be removed using existing photolithographic techniques.

[0045] FIG. 2D is a cross-sectional side view of a stage **200D** of a device of the present invention with NiO-filled interior trenches **214A-214C** and an NiO-filled peripheral trench **215** (opposing portions of which are shown at **215A** and **215B**). As noted above, the number of interior trenches is not meant to be limited to the examples herein.

[0046] With reference back to FIG. 1, at **104**, at least one junction termination extension (JTE) structure **217** is deposited about the periphery of the Ga.sub.2O.sub.3 base **202**. In general, a JTE is a technique that improves the control of electric fields in p-n junctions. JTEs can be employed to reduce field crowding at a junction periphery where the charge in the JTE is designed to sink the critical electric field lines at breakdown.

[0047] FIG. 2E is a cross-sectional side view of a stage of a device of the present invention including a JTE **217** extending about the periphery of the device. In the embodiments of FIG. 2E, a first layer **216** of a bi-layer NiO JTE structure **217** is deposited on the Ga.sub.2O.sub.3 base **202** over portions of the peripheral NiO-filled trench **215** (e.g., portions of **215A** and **215B**) and portions of a surface of the Ga.sub.2O.sub.3 epitaxial layer **204**. In implementations, the first layer **216** comprises 700 nanometers (nm) of NiO sputtered in a pure argon (Ar) atmosphere, resulting in a N_A of about 3×10¹⁷ cm⁻³. In embodiments, the first layer **216** is configured to extend beyond (e.g., ~40 micrometers beyond) a respective edge of an anode contact **220**. See FIG. 2E discussed below, for example.

[0048] In this first embodiment, a second layer **218** of the bi-layer NiO JTE structure **217** is then deposited over the first layer **216** and portions of the peripheral NiO-filled trench **215** (see portions of **215A**, **215B** covered by the bi-layer NiO JTE structure **217**. In implementations, the second layer **218** is deposited in two stages.

[0049] FIG. 2F is a close-up partial view of FIG. 2F, including the two stages of the second bi-layer of the NiO JTE structure. Stage one is depicted in FIG. 2F at **219A** and stage two is depicted in FIG. 2F at **219B**, for example. In embodiments, the first stage comprises 250 nm of NiO in a 20:1 argon:oxygen (Ar:O₂) atmosphere followed by a second stage comprising 100 nm of NiO in a 2:1 Ar:O₂ atmosphere, resulting in an N_A>10¹⁹ cm⁻³.

[0050] FIG. 2G is a cross-sectional side view of an alternative embodiment **200G** of a device of the present invention including a single layer JTE **217'**. In implementations, the JTE structure **217'** comprises NiO deposited about the periphery of the Ga.sub.2O.sub.3 base **202**. Opposing side portions **217A'** and **217B'** of the JTE **217'** are visible in FIG. 2G. More specifically, the JTE structure **217'** is deposited about the periphery of the Ga.sub.2O.sub.3 base **202** over portions of the peripheral NiO-filled trench **215** and portions of a surface of the Ga.sub.2O.sub.3 epitaxial layer **204**.

[0051] FIG. 2H is a top view of the embodiments **200G** of FIG. 2G. Note the JTE structure **217'** extends about the periphery of the semiconductor base **202**.

[0052] With reference back to FIG. 1, at **105**, an anode **220** (e.g., a platinum oxide (PtO_x) and

gold (Au) anode) is deposited over the interior NiO-filled trenches **214A-214C**, over portions of the NiO-filled peripheral trench **215**, and over portions of the JTE (e.g., over at least a portion of the second layer **218** of the NiO JTE **217**), via reactive ion sputtering. See FIGS. **2E** and **2F**. See also the embodiment of FIG. **2G**, wherein an anode **220'** is deposited over the interior NiO-filled trenches **214A-214C**, over portions of the NiO-filled peripheral trench **215**, and over portions of the JTE **217'**. In implementations, the sputtering occurs from a platinum target at 3 mTorr, 50 W in a 10:4 Ar:O₂ atmosphere.

[0053] Returning to FIG. **1**, at **106**, a backside ohmic contact **222** (e.g., a nickel (Ni) and gold (Au) contact) is deposited onto a surface of the Ga₂O₃ substrate **203** via reactive ion sputtering, resulting in a Ga₂O₃—NiO heterojunction device **200E**. In implementations, the sputtering occurs from a platinum target at 3 mTorr, 50W in a 10:4 Ar:O₂ atmosphere. See FIGS. **2E**, **2F** and **2G** discussed below.

[0054] At **107**, the Ga₂O₃—NiO heterojunction device (e.g., **200E** or **200G**) is annealed. In implementations, the annealing is performed at 275° C. in nitrogen gas (N₂) for 15 minutes.

Experimental Results

[0055] Fabrication of a PtO_x-anode MPS device including Ni-anode JBS and reference PtO_x and Ni anode SBDs was carried out on a Ga₂O₃ base comprised of a (001) oriented Sn-doped Ga₂O₃ substrate with a 5 μm thick halide vapor phase epitaxial (HVPE) epilayer with an unintentional Si doping concentration of 5×10¹⁶ moles per cubic centimeter (cm⁻³). The fabrication process began with a 100 nm PECVD SiO₂ mask deposited onto the Ga₂O₃ base and patterned using buffered oxide etchant (BOE). Atomic Ga flux etching was performed for 159 minutes (min) in a molecular beam epitaxy (MBE) reactor using a Ga flux at a pressure of ~1.24×10⁻⁷ Torr and at thermocouple temperature of 725 degrees Celsius (°C.). Prior to Ga-flux etching, the SiO₂ patterned Ga₂O₃ surface was heat cleaned for 10 min at 800° C. The MBE reactor pressure was maintained at 1×10⁻⁹ Torr during the etching. BOE was used to remove the SiO₂ mask that remained after etching, followed by a 2 min soak in dilute 10:1 H₂O:HCl to remove remaining Ga droplets from the surface of the Ga₂O₃ base. Profilometry using a scanning electron microscopy (SEM) yielded an etch depth of ~240 nm for an etch rate of ~1.51 nm/min. After re-patterning for NiO liftoff, ~240 nm of NiO with a hole concentration N_A>10¹⁸ cm⁻³ was sputtered from a NiO target into the etched trenches. The room temperature sputtering conditions were 20:1 Ar:O₂, at 100 W power, and at a 3 mTorr pressure. Subsequently, a bi-layer NiO junction termination extension (JTE) was sputtered around the devices extending to ~40 μm beyond the edge of an anode contact position.

[0056] The first JTE layer consisted of 700 nm of NiO sputtered in a pure Ar atmosphere, resulting in hole concentration N_A of about 3×10¹⁷ cm⁻³. The second JTE layer was sputtered in two parts consisting of 250 nm of 20:1 Ar:O₂ NiO followed by 100 nm of 2:1 Ar:O₂ NiO with N_A>10¹⁹ cm⁻³. All sputtering conditions, including the sputtered PtO_x anode layer are listed in Table 1 below.

TABLE-US-00001 TABLE 1 Sputtering Conditions Thickness Pressure Ar/O₂ (nm) (mTorr) Power (W) N_A (cm⁻³) JBS NiO 20:1 240 3 100 >10¹⁸ JTE 1 60:0 700 3 100 ~3 × 10¹⁷ JTE2 20:1 350 3 100 >10¹⁸ 2:1 110 3 100 >10¹⁹ PtO_x 10:4 20 3 50 —

[0057] A schematic of the bi-layer JTE and the etched trench geometry is shown in FIG. **2E**. The Ni/Au and PtO_x/Au anodes were deposited via electron beam evaporation and reactive ion sputtering, respectively. Platinum oxide was sputtered from a platinum target (3 mTorr, 50 W) in a 10:4 Ar:O₂ atmosphere. Backside Ohmic contacts of Ti/Au (20/200 nm) were deposited using electron beam evaporation. The anode area was roughly 100×100 μm with rounded corners of radius 10 μm. A backside Ohmic contact anneal was not performed. Lastly, an anneal of the as-fabricated device structures was performed at 275° C. in N₂ for 15 min.

[0058] FIG. **3A** is a chart **300A** depicting C-V measurements of a PtO_x and Ni SBD in

accordance with embodiments of the invention. Capacitance-voltage (C-V) measurements were taken at 1 MHz frequency on the reference SBDs of both Ni and PtO.sub.x anodes to extract the $\Phi_{\text{B,CV}}$ as shown in FIG. 3A. The built-in potential (V_{bi}) for the Ni and PtO.sub.x anode metals was 1.17 eV and 1.87 eV, respectively. Using the extracted V_{bi} and the additional $E_{\text{C-E,F}}$, the $\Phi_{\text{B,CV}}$ was calculated to be 1.22 eV (Ni) and 1.92 eV (PtO.sub.x). [0059] FIG. 3B is a chart 300B depicting C-V measurements of a PtO.sub.x JBS diode in accordance with embodiments of the invention, showing a dual sweep at 300 kHz measured from accumulation to depletion (dashed line) and back to accumulation (solid line). The absence of hysteresis shows that charge trapping along the etched sidewalls is minimal for the plasma-free etched Ga.sub.2O.sub.3 NiO heterojunction. As the JBS and MPS diodes were fabricated using identical mask designs, the differentiation between the two diode architectures stems from the Ohmic contact formation to the p+ NiO regions. The forward bias operation of a JBS diodes is determined mainly by majority carrier injection from the Schottky anode into the n-drift region. Since minority carrier injection from the p-region is not required, the anode does not need to form an Ohmic contact to the p-type NiO in a JBS diode. An improved Ohmic contact to the p-type region is necessary for the formation of a MPS diode where a second turn-on and increase in current density due to minority carrier injection occurs as the PN diode portion of the device turns on. In this work, the Ni and PtO.sub.x Ohmic contacts to p-NiO were evaluated using linear transmission line measurements (LTLM).

[0060] FIG. 3C is a chart 300C depicting linear transmission line measurements of Ni and PtO.sub.x Ohmic contacts to NiO. Control films of NiO were isolated via lift-off on sapphire substrates, as indicated in FIG. 3C. A specific contact resistance ($R_{\text{c,sp}}$) of $1.07 \times 10^{-2} \Omega \cdot \text{cm}^2$ was extracted from the Ni/NiO LTLMs. The PtO.sub.x/NiO LTLMs resulted in a $R_{\text{c,sp}}$ of $1.87 \times 10^{-4} \Omega \cdot \text{cm}^2$, a reduction of nearly two orders of magnitude. Similar sheet resistance values of $1.38 \times 10^6 \Omega/\text{sq}$ (Ni) and $1.46 \times 10^6 \Omega/\text{sq}$ (PtO.sub.x) were extracted as evidenced by the nearly linear slope of the LTLM data in FIG. 3C.

[0061] FIG. 4A is a chart 400A depicting C-V measurements for a PtO.sub.x MPS and Ni JBS diodes in accordance with embodiments of the invention. The forward current density voltage (J-V) characteristics were recorded using a Keithley 4200SCS source measurement unit. The JBS and MPS diodes can be analyzed by splitting the J-V characteristics into two regimes, a Schottky barrier regime ($V_{\text{F}} < 3\text{V}$) and a PN diode regime ($V_{\text{F}} > 3\text{V}$). The V_{F} value of 3 V is chosen due to the band-alignment of the Ga.sub.2O.sub.3 NiO heterojunction. The turn-on voltage (V_{on}) of the Ni/Au JBS diode was 1.13 V as shown in FIG. 4A. The minimum differential specific on-resistance ($R_{\text{on,sp}}$) of $3.12 \text{ m}\Omega \cdot \text{cm}^2$ occurred at V_{F} of 1.84 V. The differential $R_{\text{on,sp}}$ showed a gradual increase as the JBS diode entered the PN regime where a minimum $R_{\text{on,sp}}$ of $3.15 \text{ m}\Omega \cdot \text{cm}^2$ occurred at V_{F} of 5.18 V. The PtO.sub.x/Au MPS diode exhibits differing J-V/ $R_{\text{on,sp}}$ characteristics as shown in FIG. 4A. As a result of the increased PB of the PtO.sub.x anode compared to Ni, the PtO.sub.x MPS diode shows an increased V_{on} of 1.74 V. As the MPS diode turns on in the Schottky regime, the differential $R_{\text{on,sp}}$ continuously decreases unlike the Ni JBS diode. A more drastic decrease of the $R_{\text{on,sp}}$ is observed within the PN regime, where minimum value of $1.97 \text{ m}\Omega \cdot \text{cm}^2$ was observed at 3.7 V. It is at this point the improved Ohmic contact to PtO.sub.x/NiO results in a second turn-on and a higher current density compared to the Ni JBS diode. Increased forward current under surge conditions is a fundamental characteristic of a MPS diode due to minority carrier injection in these devices. Such behavior has not been reported for Ga.sub.2O.sub.3 devices before.

[0062] FIG. 4B is a chart 400B depicting C-V measurements Ni/Au and PtO.sub.x/Au SBDs in accordance with embodiments of the invention, showing a dual sweep at 300 kHz measured from accumulation to depletion and back to accumulation. Ni/Au and PtO.sub.x/Au SBDs were measured for comparison and exhibit a V_{on} of 0.97 V and 1.62 V, respectively, as depicted in

FIG. 4B. The minimum differential $R_{\text{sub.on,sp}}$ of the Ni/Au SBD is $1.54 \text{ m}\Omega\cdot\text{cm}^2$ and increases beyond this point, similar to Schottky regime of the Ni/Au JBS diode. The PtO.sub.x/Au SBD has minimum differential $R_{\text{sub.on,sp}}$ of $1.67 \text{ m}\Omega\cdot\text{cm}^2$ with a less gradual increase beyond this point.

[0063] FIG. 5 illustrates a testing waveform **500** for high voltage reliability measurements in accordance with embodiments of the invention. Reverse I-V characteristics were measured in Fluorinert (FC70) using a Keithly 2657 as shown in FIG. 5. The PtO.sub.x MPS diode achieved a breakdown voltage ($V_{\text{sub.BR}}$) of 1 kV while exhibiting a reduction in leakage current as a result of the increased PtO.sub.x $\Phi_{\text{sub.B}}$ compared to Ni/Au JBS devices. A significant increase in leakage current was observed for the Ni/Au JBS devices, particularly at lower voltages, while achieving a $V_{\text{sub.BR}}$ of 900 V. The leakage current of the PtO.sub.x/Au SBD was comparable or slightly lower than the Ni/Au JBS devices up to 600 V. Current compliance of $100 \mu\text{A}$ was reached at 800 V for the PtO.sub.x SBD. The geometry or cell pitch, along with the abrupt junction doping concentrations of a JBS and MPS diode heavily influence the leakage current. The $6 \mu\text{m}$ pitch ($4 \mu\text{m}$ Schottky and $2 \mu\text{m}$ Ohmic) of the JBS and MPS diodes is not ideal for this NiO Ga.sub.2O.sub.3 junction doping of $N_{\text{sub.A}} \sim 10^{18} \text{ cm}^{-3}$ and $N_{\text{sub.D}} = 5 \times 10^{18} \text{ cm}^{-3}$, respectively. Further optimization of the cell pitch and geometry, as well as an increase in the abrupt junction doping profile could improve the leakage current without impacting the forward characteristics of the device.

[0064] A key advantage provided by the MPS (and JBS) diode structure, in accordance with embodiments of the invention, is the reduction of leakage current at the Schottky metal-semiconductor interface, a weak point of the device. Silvaco TCAD simulations for the PtO.sub.x MPS diode and SBDs were performed at an 800 V reverse bias. The peak electric field ($E_{\text{sub.C}}$) 100 nm below the semiconductor interface showed a value of 3.22 MV/cm and 3.63 MV/cm for the MPS and SBDs, respectively. While only 0.4 MV/cm, the simulated reduction in peak $E_{\text{sub.C}}$ from the drift layer depletion was validated by the improved $V_{\text{sub.BR}}$ and leakage current from the experimental reverse I-V data.

[0065] Lastly, high bias reliability and stress testing were performed on the PtO.sub.x MPS diodes in accordance with embodiments of the invention, using a Keysight B1505. High bias reliability testing of Ga.sub.2O.sub.3 devices has not been reported in the literature to date and is a crucial test for evaluating the ruggedness and switching capabilities of Ga.sub.2O.sub.3. The testing waveform of the MPS diode according to embodiments of the invention is shown in FIG. 5. Device forward characteristics were initially measured before applying a maximum 800 V reverse bias for a cumulative stress time of 2000 s. Throughout the duration of the test, devices were periodically switched on a total of 25 times.

[0066] FIG. 6 is a chart **600** depicting specific on-resistance and reverse leakage current as a function of stress time measured at 600, 700 and 800 V. During the test, each time the MPS diode was periodically switched on, the minimum differential $R_{\text{sub.on,sp}}$ was recorded within the Schottky and PN regime. Analysis of the $R_{\text{sub.on,sp}}$ values shows minimal variation ($\sim 0.1 \text{ m}\Omega\cdot\text{cm}^2$), remaining nearly constant throughout the duration of each 2000 s measurement.

[0067] FIG. 7 is a chart **700** depicting forward J-V measurements (25 curves) taken during the reliability measurements of the MPS diode at 700V. The vertical dashed line separates the Schottky and PN regimes of the MPS diode. The forward J-V characteristics are plotted for each device turn-on that occurred during the 700 V reliability test, as depicted in FIG. 7. The nearly identical $V_{\text{sub.on}}$ and $R_{\text{sub.on,sp}}$ within the Schottky regime of the MPS diode further supports a lack of charge trapping at the plasma-etch-free interface contributing to excellent diode stability.

[0068] In conclusion, fabrication of Ga.sub.2O.sub.3 NiO heterojunction devices using a plasma-etch-free Ga flux etch method was demonstrated. MPS and JBS diodes were fabricated utilizing PtO.sub.x and Ni contacts, respectively. An increased metal work function provides a larger

Schottky barrier height to the n-type Ga.sub.2O.sub.3 for improved leakage current while simultaneously improving the Ohmic contact to p-type NiO. Analysis of the NiO Ohmic contacts shows that PtO.sub.x has a lower R.sub.c,sp compared to Ni. The improved contact resistance results in a significant reduction of the R.sub.on,sp and an increase of the current density as the PN diode portion of the MPS diode of the present invention is switched on, a trait not exhibited by the Ni JBS diodes.

[0069] The PtO.sub.x MPS diodes fabricated in accordance with embodiments of the invention achieved a V.sub.BR of 1 kV while also withstanding 800 V reliability testing for a cumulative time of 2000 s. The quality of the etched surface and lack of charge trapping at the heterojunction is supported by the nearly constant turn-on characteristics of the MPS diodes.

[0070] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A method of making a gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device comprising: providing a Ga.sub.2O.sub.3 base including a Ga.sub.2O.sub.3 substrate with an n-type Ga.sub.2O.sub.3 epitaxial layer thereon; forming NiO-filled internal trenches and an NiO-filled peripheral trench in the n-type Ga.sub.2O.sub.3 epitaxial layer using a plasma-free etching process; forming at least one junction termination extension (JTE) structure about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer; depositing an anode over the NiO filled interior trenches, over portions of the NiO-filled peripheral trench, and over portions of the JTE structure; and depositing a cathode over a bottom surface of the Ga.sub.2O.sub.3 substrate, thereby forming the Ga.sub.2O.sub.3—NiO heterojunction device, wherein the Ga.sub.2O.sub.3—NiO heterojunction device is formed without the use of plasma-etching and is free of plasma-etching damage.
2. The method of claim 1, wherein forming the NiO-filled internal trenches and an NiO-filled peripheral trench includes: forming internal trenches in the n-type Ga.sub.2O.sub.3 epitaxial layer via the plasma-free etching process; forming a peripheral trench extending about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer via the plasma-free etching process; depositing p-type NiO into the internal trenches and the peripheral trench, thereby forming the NiO-filled internal trenches and an NiO-filled peripheral trench.
3. The method of claim 1, wherein forming the JTE structure comprises: depositing the at least one JTE structure over portions of the NiO filled peripheral trench and surface portions of the Ga.sub.2O.sub.3 epitaxial layer.
4. The method of claim 1, wherein the Ga.sub.2O.sub.3—NiO heterojunction device comprises a merged PiN Schottky (MPS) diode having a specific contact resistance between the anode and the NiO of less than 0.0005 Ohm*cm.sup.2.
5. The method of claim 1, wherein the JTE structure comprises a bi-layer JTE structure including first and second layers of NiO.
6. The method of claim 5, wherein depositing the JTE structure comprises: depositing a first layer of NiO via sputtering in a pure argon atmosphere over portions of the peripheral NiO-filled trench and over portions of a surface of the Ga.sub.2O.sub.3 epitaxial layer; and depositing a first stage of a second layer of NiO in 20:1 argon:oxygen atmosphere over the first layer and portions of the peripheral NiO-filled trench; and depositing a second stage of the second layer of NiO in a 2:1

argon:oxygen atmosphere over the first stage of the second layer of NiO and portions of the peripheral NiO-filled trench.

7. The method of claim 1, wherein the JTE structure comprises a layer of NiO.

8. The method of claim 1, wherein forming the NiO-filled internal trenches and the NiO-filled peripheral trench in the n-type Ga.sub.2O.sub.3 epitaxial layer using a plasma-free etching process comprises patterning using photolithography and gallium flux etching.

9. The method of claim 1, further comprising annealing the Ga.sub.2O.sub.3—NiO heterojunction device.

10. The method of claim 1, wherein the anode and cathode comprise platinum oxide and gold.

11. The method of claim 1, wherein the n-type Ga.sub.2O.sub.3 epitaxial layer is doped with silicon (Si).

12. A gallium oxide (Ga.sub.2O.sub.3)-nickel oxide (NiO) heterojunction device comprising: a Ga.sub.2O.sub.3 base comprised of a Ga.sub.2O.sub.3 substrate with an n-type Ga.sub.2O.sub.3 epitaxial layer thereon, the n-type Ga.sub.2O.sub.3 epitaxial layer including a plurality of NiO-filled internal trenches and an NiO-filled peripheral trench; a junction termination extension (JTE) structure formed about the periphery of the n-type Ga.sub.2O.sub.3 epitaxial layer; an anode formed over the NiO filled interior trenches, over portions of the NiO-filled peripheral trench, and over portions of the JTE structure; and a cathode formed over a bottom surface of the Ga.sub.2O.sub.3 substrate, wherein the Ga.sub.2O.sub.3—NiO heterojunction device is formed without the use of plasma-etching and is free of plasma-etching damage.

13. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the JTE structure is a NiO bi-layer JTE structure.

14. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 13, wherein the first layer of the NiO bi-layer JTE structure extends beyond opposing edges of the anode.

15. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the JTE structure comprises a layer of NiO.

16. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the anode and cathode comprise platinum oxide and gold.

17. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the JTE structure is formed over portions of the NiO filled peripheral trench and surface portions of the Ga.sub.2O.sub.3 epitaxial layer.

18. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the Ga.sub.2O.sub.3—NiO heterojunction device comprises a merged PiN Schottky (MPS) diode having a specific contact resistance between the anode and the NiO of less than 0.0005 Ohm*cm.sup.2.

19. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the n-type Ga.sub.2O.sub.3 epitaxial layer is doped with silicon (Si).

20. The Ga.sub.2O.sub.3—NiO heterojunction device of claim 12, wherein the n-type Ga.sub.2O.sub.3 epitaxial layer has a thickness of at least 5 microns.
